

Question: 1

Draw a network topology diagram for a small café Wi-Fi setup using the star topology, labeling all devices (router, access points, user devices) and connections clearly?

- Answer:
1. Router is the central device of the network.
 2. Access Points (APs) are connected to the router.
 3. devices like laptops, mobiles, and tablets connect to the access points.
 4. All devices communicate through the central router/access point.
 5. The connections form a star topology, where devices are connected to a central point.
 6. Wi-Fi is used to connect the user devices wirelessly.
 7. If one user device disconnects, the other devices continue to work.

Question: 2

Design a network for a two-floor co-working space (like a mini WeWork) using a ring topology for each floor and explain how devices are connected to maintain communication between floors.
Hint: Show how the ring topology helps in handling device failures on one floor without affecting the other?

- Answer:
1. Each floor has a ring topology where all devices are connected in a circular manner.
 2. Each floor has a switch/router to manage the network.
 3. The two floors are connected through a main network link.
 4. Devices on the same floor communicate through the ring.

4. Data can travel around the ring to reach the required device.
5. If one device on one floor fails, the other floor can continue working normally.
6. The ring topology helps reduce the effect of a device failure on other floors.
7. Proper backup links can be used to maintain communication if a connection breaks.

Question: 3

Create a mesh topology diagram for a gaming café with 6 PCs, showing how redundancy is achieved if any one PC or cable fails?

Answer: 1. A mesh topology connects 6 PCs directly with multiple network connections.

2. Each PC is connected to the other PCs through separate cables.
3. This provides multiple paths for data transmission.
4. If one PC fails, the other PCs can continue communicating.
5. If one cable fails, data can travel through another available path.
6. Mesh topology provides high reliability and redundancy.
7. It requires more cables and is expensive to install and maintain.

Question: 4

Plan a multi-site network for a food delivery company with 3 city offices (Ahmedabad, Surat, Vadodara) — show how you would connect the sites, what topology you would use between them, and briefly explain your choice?

Answer: 1. Connect the three offices: Ahmedabad, Surat, and Vadodara using a WAN.

2. Use a VPN/leased line to securely connect all three offices.

3. Use a mesh topology between the three city offices.

4. Each office will have a router, switch, computers, and server.

5. Ahmedabad can connect directly to Surat and Vadodara.

6. Surat can connect directly to Ahmedabad and Vadodara.

7. Vadodara can connect directly to Ahmedabad and Surat.

8. Mesh topology provides reliable communication and backup paths if one connection fails.

9. This topology is suitable for a multi-site company because it provides better availability and security.

Question: 5

Imagine your network needs to double in size next year (e.g., from 10 to 20 devices per site). Write a short expansion plan describing what changes you would make to your original topology to ensure scalability and minimal downtime.

Hint: Consider adding switches, routers, or changing topology types if needed.?

Answer: 1. Upgrade the existing switches and routers to support more devices.

2. Add additional switches when the number of devices increases.

3. Use a star or hierarchical star topology for easy expansion.

4. Increase the IP address range to support more devices.

5. backup routers and connections to reduce downtime.

6. Use high-speed cables and network equipment for better performance.

7. Monitor the network regularly to identify and fix problems quickly.
8. Expand the network gradually from 10 to 20 devices per site without changing the complete network.